

VeriTrust AI — Technical Specification Dossier

Enterprise AI Governance, Multi-Tier Evidence Retrieval & Deterministic Risk Engine

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Target Domain Exposure: BFSI / Financial Services, Healthcare & Life Sciences, HR & Employment, Aviation & Aerospace etc.

1. Production Deployment & Live Infrastructure Endpoints

The platform is engineered as a decoupled, high-performance distributed cloud system:

Component	Cloud Provider	Live Production Endpoint	Operational Status
Frontend SPA UI	Vercel Edge Network	https://veri-trust-ai-governance.vercel.app	■ 100% Operational
Backend REST Engine	Render Web Service	https://veritrust-ai-governance-1.onrender.com	■ 100% Operational
Interactive OpenAPI Docs	FastAPI Swagger UI	https://veritrust-ai-governance-1.onrender.com/docs	■ 100% Operational
API Health Probe	Render Microservice	https://veritrust-ai-governance-1.onrender.com/api/health	■ 100% Operational
Source Code Repository	GitHub Public Repo	https://github.com/vanshika-data-lab/VeriTrust-AI-Governance	■ Main Branch Synced

2. Mandatory 5-Layer Enterprise AI System Architecture

The system strictly implements the Modus 5-Layer Enterprise AI Architecture specification:

Layer	Technical Components	Deep Implementation & Engineering Responsibilities
Layer 1: User Interface	React 18.3, Vite 5.4, Lucide-React, Vanilla CSS Tokens	Executive KPI overview, 10-area dynamic visual risk meters, surprise record interactive form, 6-tier knowledge explorer, detailed audit report viewer, and instant JSON compliance export.
Layer 2: Application / API	FastAPI 0.110+, Uvicorn 0.28+, Pydantic v2, CORS Middleware	Stateless asynchronous REST routing, strict request/response schema validation, automated Swagger/OpenAPI v3 generation, and error-handling middleware with cold-start detection.
Layer 3: AI Intelligence	Python 3.10+, scoring_matrix.py, ai_synthesis.py	Mathematical 10-dimension deterministic risk scoring matrix, weighted aggregation, EU AI Act conformity classification, and statutory obligation/mitigating control reasoning.
Layer 4: Data & Knowledge	SQLite3, db_manager.py, Relational Store	ACID-compliant persistence across 5 relational tables (use_cases, assessments, dimension_assessments, evidence_sources, knowledge_base) with foreign key cascade deletions.
Layer 5: External Research	HTTPX Async, research_retriever.py, evidence_classifier.py	Hybrid RAG engine: query inverted indexing over 15 pre-seeded canonical regulatory statutes + live asynchronous HTTP web crawling for novel evaluator AI systems.

3. Deterministic 10-Dimension Risk Scoring Mathematics

Unlike opaque black-box LLM prompt scoring, VeriTrust AI implements a repeatable, mathematically rigorous scoring formula across 10 governance areas:

Mathematical Formulation:

1. Overall Risk Index: $R_{\text{overall}} = (\sum (w_i \cdot S_i)) / (\sum w_i)$ where w_i is dimension weight and $S_i \in [10, 98]$.
2. Data Sensitivity Factor: $f_{\text{data}} = \min(1.0, 0.3 + 0.2 \cdot N_{\text{sensitive}})$ where $N_{\text{sensitive}}$ is count of sensitive data types.
3. Autonomy Multiplier (α): Fully Autonomous = 1.00, Human-on-the-Loop = 0.75, Human-in-the-Loop = 0.50, Advisory = 0.30.
4. Impact Multiplier (β): Critical = 1.00, High = 0.80, Medium = 0.50, Low = 0.20.
5. Population Scale Multiplier (γ): >500k = 1.00, >50k = 0.85, >5k = 0.70, <=5k = 0.50.

Dimension Key	Dimension Name	Weight	Deterministic Computation Formula	Canonical Statutory Mandate
data	Data Lineage & Quality	1.0 (8.1%)	$Base(40) + 35 \cdot f_{\text{data}} + 25 \cdot \beta_{\text{impact}}$	EU AI Act Art. 10 / ISO 42001 Annex A.8
privacy	Privacy & Data Protection	1.2 (9.8%)	$Base(40) + 40 \cdot f_{\text{data}} + 25 \cdot Ind_{\text{priv}}$	GDPR Art. 35 DPIA / FTC Fair Info
bias_fairness	Bias, Fairness & Equity	1.3 (10.6%)	$Base(40) + 35 \cdot Ind_{\text{bias}} + 25 \cdot \beta_{\text{impact}}$	NYC LL144 / Title VII 80% Rule / ECOA
human_oversight	Human Oversight Controls	1.4 (11.4%)	$Base(40) + 45 \cdot \alpha_{\text{autonomy}} + 15 \cdot \beta_{\text{impact}}$	EU AI Act Art. 14 / NIST GOVERN 1.2
explainability	Explainability & Recourse	1.1 (8.9%)	$Base(40) + 30 \cdot \alpha_{\text{autonomy}} + 30 \cdot \beta_{\text{impact}}$	EOCA 12 CFR 1002.9 / GDPR Art. 22
security	Cybersecurity Robustness	1.2 (9.8%)	$Base(40) + 30 \cdot \gamma_{\text{scale}} + 30 \cdot \alpha_{\text{autonomy}}$	SOC 2 CC6.1 / NIST AI RMF Manage
decision_impact	Decision Impact Severity	1.5 (12.2%)	$Base(40) + 50 \cdot \beta_{\text{impact}} + 10 \cdot \gamma_{\text{scale}}$	ISO/IEC 23894 / Microsoft RAIA Std
regulatory_exposure	Regulatory Exposure	1.4 (11.4%)	$Base(40) + 40 \cdot Ind_{\text{reg}} + 20 \cdot \beta_{\text{impact}}$	EU AI Act High-Risk Annex III
model_risk	Model Reliability & Drift	1.0 (8.1%)	$Base(40) + 35 \cdot \alpha_{\text{autonomy}} + 25 \cdot \gamma_{\text{scale}}$	Stanford HAI / NIST MEASURE 2.2
monitoring	Continuous Monitoring	1.0 (8.1%)	$Base(40) + 30 \cdot \gamma_{\text{scale}} + 30 \cdot \alpha_{\text{autonomy}}$	ISO/IEC 42001 Clause 6.1 Telemetry

4. 6-Tier Regulatory Evidence Classification & RAG Retriever

The retrieval engine implements a strict hierarchical source weighting algorithm (evidence_classifier.py) across 6 distinct tiers:

Source Tier	Reliability Weight	Heuristic Classification Criteria & Keywords	Pre-Seeded Canonical Reference Authorities
Tier 1: Law / Regulation	0.98 (Highest)	Act, Regulation, Statute, Directive, CFR, Law, GDPR, Article	EU AI Act (2024/1689), GDPR (2016/679), NYC LL144, ECOA Reg B
Tier 2: Regulatory Guidance	0.93	Guidance, Framework, Circular, Advisory, Bulletin, Enforcement	NIST AI RMF 1.0, FTC Fair Information Circular, EEOC Guidance
Tier 3: Industry Standard	0.90	ISO, IEC, IEEE, SOC 2, AICPA, Standard, ANSI, BSI	ISO/IEC 42001:2023 (AIMS), IEEE 7000-2021, Microsoft RAIA v2
Tier 4: Academic Research	0.85	Paper, Journal, Proceedings, ACM, ArXiv, Empirical Study, DOI	Stanford CRFM Benchmarks, MIT Disparate Impact Audits
Tier 5: Vendor Information	0.78	System Card, Cloud SLA, Model Card, API Documentation, Vendor	Foundation Model System Cards, Cloud AI SOC 2 Compliance Audits
Tier 6: General Web Content	0.65 (Baseline)	Real-time public web crawling via HTTPX client for novel use cases	Audited Industry Regulatory Trackers, Public Legal Repositories

5. Relational Database Schema & Data Dictionary (SQLite)

The persistence layer uses a normalized SQLite relational architecture (backend/database/governance_app.db):

Table Name	Primary Key	Foreign Key Relationships	Attributes & Column Types	Storage Semantics
use_cases	id (INTEGER)	Root entity (Parent)	name, industry, purpose, autonomy_level, data_types (JSON), affected_population, impact_tier, is_preseeded, created_at	Stores AI system architecture specifications and operational profiles.
assessments	id (INTEGER)	use_case_id -> use_cases.id (CASCADE)	use_case_id, overall_risk_score, risk_level, eu_ai_act_category, executive_summary, created_at	Stores aggregate assessment scores and EU AI Act regulatory tiering.
dimension_assessments	id (INTEGER)	assessment_id -> assessments.id (CASCADE)	assessment_id, dimension_key, dimension_name, risk_score, risk_level, findings, regulatory_impact, mitigating_controls (JSON)	Maintains exactly 10 dimension rows per assessment with concrete controls.
evidence_sources	id (INTEGER)	assessment_id -> assessments.id (CASCADE)	assessment_id, dimension_key, source_tier, title, author_entity, citation_text, url, jurisdiction, pub_date, reliability_score	Stores 6-tier retrieved statutory citations supporting each finding.
knowledge_base	id (INTEGER)	Independent reference store	source_tier, title, author_entity, jurisdiction, pub_date, url, summary_content, key_rules (JSON), tags	Pre-loaded with 15 canonical statutes and standards across all 6 tiers.

6. REST API Specification & Endpoint Lifecycle

The backend exposes a stateless, high-performance REST API built on FastAPI and Uvicorn:

HTTP Method	Endpoint Route	Request Payload Schema	Response Status & Data Model
GET	/api/health	None	200 OK: { status: 'healthy', service: 'AegisAI Governance Engine' }
GET	/api/use-cases	None	200 OK: Array of all evaluated AI use cases with risk scores.
DELETE	/api/use-cases/{id}	Path param: id (int)	200 OK: Cascading deletion of use case, assessments, & evidence.
GET	/api/assessments/{id}	Path param: id (int)	200 OK: Full assessment object with 10 dimensions & 6-tier citations.
POST	/api/assess	JSON: { name, industry, purpose, autonomy_level, data_types, affected_population, impact_tier }	200 OK: Evaluates novel surprise use case, computes scores, retrieves RAG citations, & persists.
GET	/api/sources	Query params: ?query=...&source_tier=...	200 OK: Filtered knowledge base articles with statutory mandates.
GET	/api/analytics	None	200 OK: Aggregate metrics, tier distribution, & industry breakdown.
GET	/api/export-report/{id}	Path param: id (int)	200 OK: Complete compliance audit report JSON download payload.

7. Open-Source Technology Stack & Resilience Strategy

All components comply 100% with the Modus challenge rules requiring zero paid software licenses:

Library / Tool	Version	Open-Source License	Purpose in Platform & Resilience Strategy
Python	3.10+	PSFL (Open Source)	Core backend runtime environment.
FastAPI	0.110.0+	MIT License	High-speed asynchronous REST API framework with native Pydantic validation.
Uvicorn	0.28.0+	BSD-3-Clause	Production ASGI web server supporting async coroutines.
SQLite3	3.x	Public Domain / Free	Embedded, ACID-compliant relational store. Requires zero external database server.
React	18.3.1	MIT License	Component-driven reactive single-page frontend application.
Vite	5.4.21	MIT License	Frontend bundler with tree-shaking, fast HMR, and optimized production chunking.
Lucide-React	0.441.0	ISC License	Executive iconography for governance dimensions and 6 source tiers.

8. Compliance & Candidate Verification Statement

This technical dossier certifies that VeriTrust AI fulfills all requirements of the Modus Enterprise AI Build Challenge Assignment 7. The platform delivers repeatable, deterministic risk scoring across 10 governance dimensions, classifies evidence across 6 hierarchical source tiers, processes surprise records dynamically, persists data in SQLite, and is fully deployed in production on Vercel and Render.